

UCab — Full-Stack MERN Cab Booking Platform

Solo Developer / Team Member: Shaik Sumiya Zainab | Project ID: N/A (Solo Track)

Phase 7: Project Documentation — Exhaustive User Manual & API Reference

Project Name: Cab Booking (`UCab`)

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1. System Operating Manual

1.1 For Urban Commuters (Riders)

1. Navigating the Homepage (`/home`): The landing page welcomes riders with a vibrant hero section and an instant booking search widget. Enter your desired pickup point, drop-off location, and trip date, then click Find Available Cabs.
2. Filtering Fleet Inventory (`/cabs`): Browse through all available vehicles categorized by Mini (`10/km`), Sedan (`12/km`), SUV (`18/km`), and Luxury (`35/km`). Use the top filter tabs or sort by price (`Low to High`) to find a cab that fits your budget.
3. Customizing the Ride (`/book/:id`):
 - Distance Estimator: The system automatically calculates your trip distance (`km`) and estimated travel time (`minutes`).
 - Promo Discounts: Type `UCAB20` into the discount box and click Apply to deduct exactly 20% from your fare.
 - Add-On Perks: Check In-Ride Chilled Water & Snacks Package (+ `50`) or Road Safety Driver Welfare Donation (+ `20`). Notice how the total fare updates instantly.
 - Payment Selector: Choose Automatic Saved Card (`Visa \*\*\*\* 4242`) or UPI Instant Checkout, then click Confirm & Book Cab.
4. Live Dispatch Tracking (`/home-user`): Watch the interactive progress bar advance across 4 stages (`Booking Accepted` ! `Driver Assigned` ! `Cab in Transit` ! `Completed`). Inspect your assigned driver details (`Vikram Sharma - 4.9 & `, phone number, and registration plate `DL 01 AB 1234`).
5. Downloading Corporate Receipts (`/bookings`): Visit your trip log, locate any completed ride, and click Invoice (`ReceiptModal.jsx`). Print or download your formal thermal tax invoice with embedded QR Verification Code.

1.2 For Dispatch Administrators (`/admin/\*`)

1. Executive Login (`/admin/login`): Click "Ø=þÁþ One-Click Super Admin Login" to bypass manual typing during evaluation.
2. Monitoring Analytics KPIs (`/admin`): Inspect top-level cards showing Gross Revenue (`1`), Total Bookings, Active Users, and Operational Fleet Count.
3. Managing Fleet Inventory (`/admin/cars`): View all active cabs. Click Add New Cab (`/admin/add-car`) to create a new vehicle profile by entering model name, category, pricing, and attaching a high-res vehicle image. Click Edit (`/admin/edit-car/:id`) to update rates or toggle vehicle operational availability between `Available` (Green Badge) and `Busy` (Red Badge).
4. Dispatching & Updating Trips (`/admin/bookings`): Monitor incoming platform bookings in real time. Select any trip and change its status dropdown from `Accepted` to `Started` or `Completed` to update the rider's tracking screen instantly.

2. Comprehensive REST API Reference Guide

2.1 Authentication & User Endpoints (`/api/users`)

- `POST /api/users/register`

- Payload: `{ name: "John Doe", email: "john@ucab.com", password: "secretpassword", phone: "9876543210" }`
- Response (`201 Created`): `{ \_id: "user\_1", name: "John Doe", email: "john@ucab.com", token: "eyJhbGciOi..." }`
- `POST /api/users/login`
- Payload: `{ email: "sumiya@ucab.com", password: "password123" }`
- Response (`200 OK`): `{ \_id: "user\_p", name: "Shaik Sumiya Zainab", email: "sumiya@ucab.com", token: "eyJhbGciOi..." }`

2.2 Fleet Catalog Endpoints (`/api/cars`)

- `GET /api/cars`
- Access: Public
- Response (`200 OK`): Array of car objects `[ { \_id: "car\_1", name: "Maruti Suzuki Swift Dzire", cabType: "Sedan", pricePerKm: 12, baseFare: 50, isAvailable: true, image: "https://..." }, ... ]`
- `POST /api/cars`
- Access: Protected (`Bearer Token + Admin Role`)
- Payload: Multipart/form-data or JSON `{ name, cabType, pricePerKm, baseFare, seats, plateNumber, image }`
- Response (`201 Created`): Created car object.

2.3 Trip Booking Endpoints (`/api/bookings`)

- `POST /api/bookings`
- Access: Protected User (`Bearer Token`)
- Payload: `{ car: "car\_1", pickupLocation: "Connaught Place", dropLocation: "Terminal 3", bookingDate: "2026-07-15", distanceKm: 18, totalFare: 266, refreshmentsOrdered: true, donationMade: false, discountCode: "UCAB20" }`
- Response (`201 Created`): Trip object with `driverAssigned` details.
- `GET /api/bookings/user`
- Access: Protected User (`Bearer Token`)
- Response (`200 OK`): Chronological array of personal trip records.
- `PUT /api/bookings/:id`
- Access: Protected Admin (`Bearer Token + Admin Role`)
- Payload: `{ status: "Started" }` (or `Completed`, `Cancelled`)
- Response (`200 OK`): Updated booking object.

2.4 Executive Admin Endpoints (`/api/admin`)

- `POST /api/admin/login`
- Payload: `{ email: "admin@ucab.com", password: "adminpassword" }`
- Response (`200 OK`): `{ \_id: "adm\_1", name: "Super Admin", email: "admin@ucab.com", role: "admin", token: "..." }`
- `GET /api/admin/stats`
- Access: Protected Admin (`Bearer Token + Admin Role`)
- Response (`200 OK`): `{ totalCars: 6, totalBookings: 3, totalUsers: 2, totalRevenue: 850 }`